Chapter 1

Demo problem: Turek & Hron's FSI benchmark problem

In this example we consider the flow in a 2D channel past a cylinder with an attached elastic "flag". This is the FSI benchmark problem proposed by Turek & Hron,

"Proposal for Numerical Benchmarking of Fluid-Structure Interaction between an Elastic Object and a Laminar Incompressible Flow", S. Turek & J. Hron, pp. 371-385. In: "Fluid-Structure Interaction" Springer Lecture Notes in Computational Science and Engineering **53**. Ed. H.-J. Bungartz & M. Schaefer. Springer Verlag 2006.

The problem combines the two single-physics problems of

- `Flow past a cylinder with a "flag" whose motion is prescribed.`
- `The deformation of a finite-thickness cantilever beam (modelled as a 2D solid), loaded by surface tractions.`

This is our first example problem that involves the coupling between a fluid and "proper" solid (rather than beam structure) and also includes both fluid and wall inertia.

The problem presented here was used as one of the test cases for `oomph-lib's` FSI preconditioner; see

`Heil, M., Hazel, A.L. & Boyle, J. (2008): Solvers for large-displacement fluid-structure interaction problems: Segregated vs. monolithic approaches. Computational Mechanics.`

In this tutorial we concentrate on the problem formulation. The application of the preconditioner is discussed `elsewhere` – the required source code is contained in the `driver code.`

1.1 The Problem

The figure below shows a sketch of the problem: A 2D channel of height H^* and length L^* conveys fluid of density ρ_f and dynamic viscosity μ and contains a cylinder of diameter d^*, centred at (X_c^*, Y_c^*) to which a linearly elastic "flag" of thickness H_{flag}^* and length L_{flag}^* is attached. Steady Poiseuille flow with average velocity U^* is imposed at the left end of the channel while we assume the outflow to be parallel and axially traction-free. We model the flag as a linearly elastic Hookean solid with elastic modulus E^*, density ρ_s and Poisson's ratio ν.

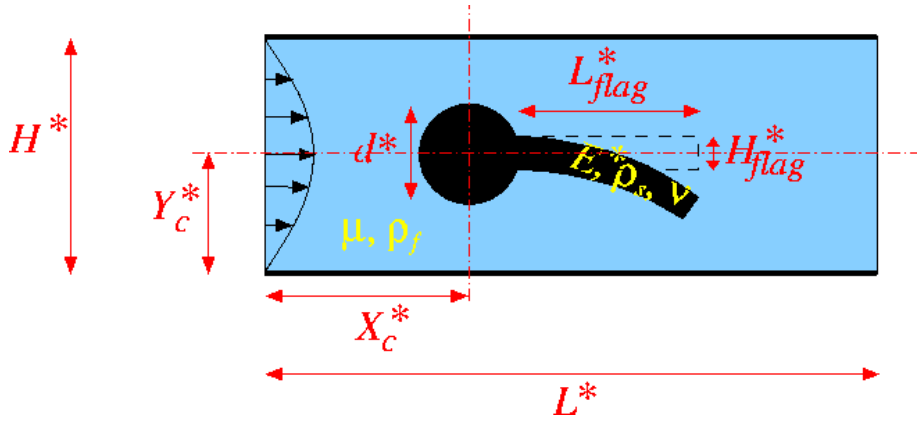

Figure 1.1 Sketch of the problem in dimensional variables.

We non-dimensionalise all length and coordinates on the diameter of the cylinder, d^*, the velocities on the mean velocity, U^*, and the fluid pressure on the viscous scale. To facilitate comparisons with Turek & Hron's dimensional benchmark data (particularly for the period of the self-excited oscillations), we use a timescale of $T^* = 1$ sec to non-dimensionalise time. The fluid flow is then governed by the non-dimensional Navier-Stokes equations

$$Re\left(St\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right],$$

where $Re = \rho U^* H_0^*/\mu$ and $St = d^*/(U^*T^*)$, and

$$\frac{\partial u_i}{\partial x_i} = 0,$$

subject to parabolic inflow

$$\mathbf{u} = 6x_2(1-x_2)\mathbf{e}_1$$

at the inflow cross-section; parallel, axially-traction-free outflow at the outlet; and no-slip on the stationary channel walls and the surface of the cylinder, $\mathbf{u} = \mathbf{0}$. The no-slip condition on the moving flag is

$$\mathbf{u} = St\,\frac{\partial \mathbf{R}_w(\xi_{[top,tip,bottom]}, t)}{\partial t} \qquad (1)$$

where $\xi_{[top,tip,bottom]}$ are Lagrangian coordinates parametrising the three faces of the flag.
We describe the deformation of the elastic flag by the non-dimensional position vector $\mathbf{R}(\xi^1, \xi^2, t)$ which is determined by the principle of virtual displacements

$$\int_v \left\{\sigma^{ij}\,\delta\gamma_{ij} - \left(\mathbf{f} - \Lambda^2\frac{\partial^2\mathbf{R}}{\partial t^2}\right)\cdot\delta\mathbf{R}\right\}\,dv - \oint_{A_{tract}} \mathbf{t}\cdot\delta\mathbf{R}\,dA = 0, \qquad (2)$$

where all solid stresses and tractions have been non-dimensionalised on Young's modulus, E^*; see the Solid Mechanics Tutorial for details. The solid mechanics timescale ratio (the ratio of the timescale T^* chosen to non-dimensionalise time, to the intrinsic timescale of the solid) can be expressed in terms of the Reynolds and Strouhal numbers, the density ratio, and the FSI interaction parameter as

$$\Lambda^2 = \left(\frac{d^*}{T^*}\sqrt{\frac{\rho_s}{E^*}}\right)^2 = St^2\left(\frac{\rho_s}{\rho_f}\right)Re\,Q.$$

Here is a sketch of the non-dimensional version of the problem:

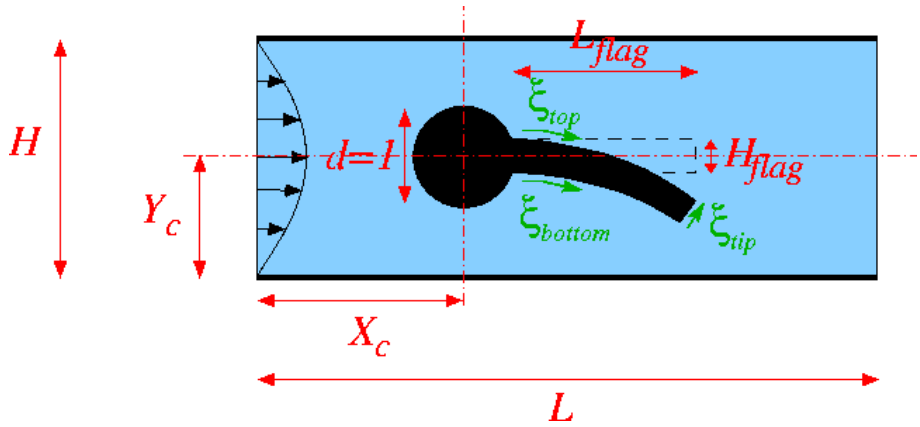

Figure 1.2 Sketch of the fluid problem in dimensionless variables, showing the Lagrangian coordinates that parametrise the three faces of the flag.

1.2 Parameter values for the benchmark problems

The (dimensional) parameter values given in Turek & Hron's benchmark correspond to the following non-dimensional parameters:

1.2.1 Geometry

- Cylinder diameter $d = 1$
- Centre of cylinder $X_c = Y_c = 2$
- Channel length $L = 25$
- Channel width $H = 4.1$
- Thickness of the undeformed flag $H_{flag} = 0.2$
- Right end of undeformed flag $x_{tip} = 6$

1.2.2 Non-dimensional parameters

The three FSI test cases correspond to the following non-dimensional parameters:

..	$Re = U^* d^* \rho_f / \mu$	$St = d^* / (U^* T^*)$	$Q = \mu U^* / (E^* d^*)$	ρ_s / ρ_f	$\Lambda^2 = (d^* / T^* \sqrt{\rho_s / E^*})^2 = St^2 (\rho_s / \rho_f) Re\, Q$
FSI1	20	0.5	1.429×10^{-6}	1	7.145×10^{-6}
FSI2	100	0.1	7.143×10^{-6}	10	7.143×10^{-6}
FSI3	200	0.05	3.571×10^{-6}	1	1.786×10^{-6}

1.3 Results

The test cases FSI2 and FSI3 are the most interesting because the system develops large-amplitude self-excited oscillations

1.3.1 FSI2

Following an initial transient period the system settles into large-amplitude self-excited oscillations during which the oscillating flag generates a regular vortex pattern that is advected along the channel. This is illustrated in the figure

below which shows a snapshot of the flow field (pressure contours and instantaneous streamlines) at $t = 6.04$.

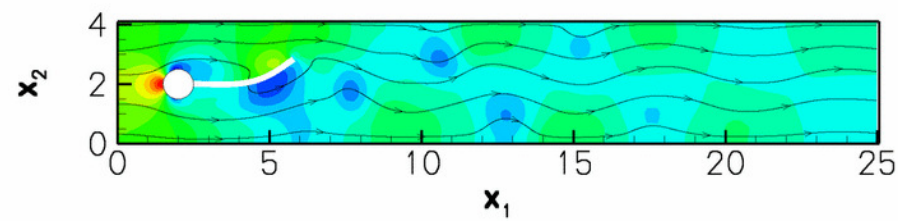

Figure 1.3 Snapshot of the flow field (instantaneous streamlines and pressure contours)

The constantly adapted mesh contains and average of 65,000 degrees of freedom. A relatively large timestep of $\Delta t = 0.01$ – corresponding to about 50 timesteps per period of the oscillation – was used in this computation. With this discretisation the system settles into oscillations with a period of ≈ 0.52 and an amplitude of the tip-displacement of 0.01 ± 0.83.

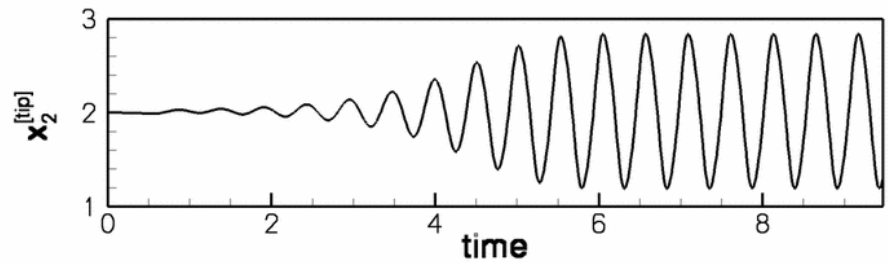

Figure 1.4 Time trace of the tip displacement.

1.3.2 FSI3

The figures below shows the corresponding results for the case FSI3 in which the fluid and solid densities are equal and the Reynolds number twice as large as in the FSI2 case. The system performs oscillations of much higher frequency and smaller amplitude. This is illustrated in the figure below which shows a snapshot of the flow field (pressure contours and instantaneous streamlines) at $t = 3.615$.

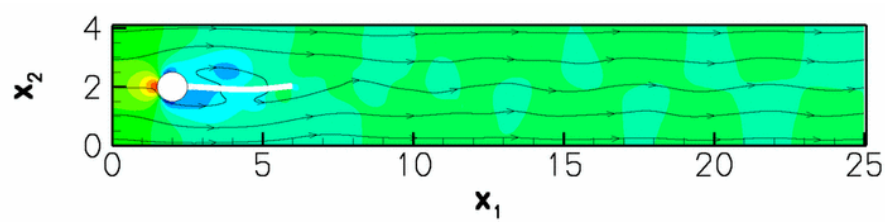

Figure 1.5 Snapshot of the flow field (instantaneous streamlines and pressure contours)

This computation was performed with a timestep of $\Delta t = 0.005$ and resulted in oscillations with a period of ≈ 0.19 and an amplitude of the tip-displacement of 0.01 ± 0.36.
The increase in frequency and Reynolds number leads to the development of thinner boundary and shear layers which require a finer spatial resolution, involving an average of 84,000 degrees of freedom.

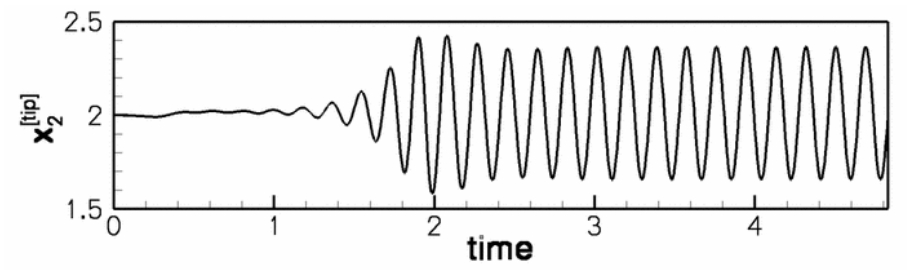

Figure 1.6 Time trace of the tip displacement.

1.4 Overview of the driver code

Since the driver code is somewhat lengthy we start by providing a brief overview of the main steps in the `Problem` construction:

1. We start by discretising the flag with 2D solid elements, as in the corresponding `single-physics solid mechanics example.`

2. Next we attach `FSISolidTractionElements` to the three solid mesh boundaries that are exposed to the fluid traction. These elements are used to compute and impose the fluid traction onto the solid elements, using the flow field from the adjacent fluid elements.

3. We now combine the three sets of `FSISolidTractionElements` into three individual (sub-)meshes and convert these to `GeomObjects`, using the `MeshAsGeomObject` class.

4. The `GeomObject` representation of the three surface meshes is then passed to the constructor of the fluid mesh. The `algebraic node-update methodology` provided in the `AlgebraicMesh` base class is used to update its nodal positions in response to the motion of its bounding `GeomObjects`.

5. Finally, we use the helper function `FSI_functions::setup_fluid_load_info_for_solid_elements(...)` to set up the fluid-structure interaction – this function determines which fluid elements are adjacent to the Gauss points in the `FSISolidTractionElements` that apply the fluid traction to the solid.

6. Done!

1.5 The global parameters

As usual, We use a namespace to define the (many) global parameters, using default assignments for the FSI1 test case.

```
//=====start_of_global_parameters==================================
/// Global variables
//=================================================================
namespace Global_Parameters
{

 /// Default case ID
 string Case_ID="FSI1";
```

```
/// Reynolds number (default assignment for FSI1 test case)
double Re=20.0;

/// Strouhal number (default assignment for FSI1 test case)
double St=0.5;

/// Product of Reynolds and Strouhal numbers (default
/// assignment for FSI1 test case)
double ReSt=10.0;

/// FSI parameter (default assignment for FSI1 test case)
double Q=1.429e-6;

/// Density ratio (solid to fluid; default assignment for FSI1
/// test case)
double Density_ratio=1.0;

/// Height of flag
double H=0.2;

/// x position of centre of cylinder
double Centre_x=2.0;

/// y position of centre of cylinder
double Centre_y=2.0;

/// Radius of cylinder
double Radius=0.5;

/// Pointer to constitutive law
ConstitutiveLaw* Constitutive_law_pt=0;

/// Timescale ratio for solid (dependent parameter
/// assigned in set_parameters())
double Lambda_sq=0.0;

/// Timestep
double Dt=0.1;

/// Ignore fluid (default assignment for FSI1 test case)
bool Ignore_fluid_loading=false;

/// Elastic modulus
double E=1.0;

/// Poisson's ratio
double Nu=0.4;
```

We also include a gravitational body force for the solid. (This is only used for the solid mechanics test cases, CSM1 and CSM2, which will not be discussed here.)

```
/// Non-dim gravity (default assignment for FSI1 test case)
double Gravity=0.0;

/// Non-dimensional gravity as body force
void gravity(const double& time,
             const Vector<double> &xi,
             Vector<double> &b)
{
 b[0]=0.0;
 b[1]=-Gravity;
}
```

The domain geometry and flow field are fairly complex and it is difficult to construct a good initial guess for the Newton iteration. To ensure its convergence at the beginning of the simulation we therefore employ the method suggested by Turek & Hron: We start the flow from rest and ramp up the inflow profile from zero to its maximum value. The parameters for the time-dependent increase in the influx are defined here:

```
/// Period for ramping up in flux
double Ramp_period=2.0;

/// Min. flux
double Min_flux=0.0;

/// Max. flux
double Max_flux=1.0;

/// Flux increases between Min_flux and Max_flux over
/// period Ramp_period
double flux(const double& t)
{
 if (t<Ramp_period)
  {
   return Min_flux+(Max_flux-Min_flux)*
    0.5*(1.0-cos(MathematicalConstants::Pi*t/Ramp_period));
  }
```

```
  else
   {
    return Max_flux;
   }
 } // end of specification of ramped influx
```

Finally, we provide a helper function that assigns the parameters for the various test cases, depending on their ID ("FSI1", "FSI2", "FSI3", "CSM1" or "CSM2"). Here is the assignment for the case FSI1:

```
 /// Set parameters for the various test cases
 void set_parameters(const string& case_id)
 {

  // Remember which case we're dealing with
  Case_ID=case_id;

  // Setup independent parameters depending on test case
  if (case_id=="FSI1")
   {
    // Reynolds number based on diameter of cylinder
    Re=20.0;

    // Strouhal number based on timescale of one second
    St=0.5;

    // Womersley number
    ReSt=Re*St;

    // FSI parameter
    Q=1.429e-6;

    // Timestep -- aiming for about 40 steps per period
    Dt=0.1;

    // Density ratio
    Density_ratio=1.0;

    // Gravity
    Gravity=0.0;

    // Max. flux
    Max_flux=1.0;

    // Ignore fluid
    Ignore_fluid_loading=false;
    // Compute dependent parameters

    // Timescale ratio for solid
    Lambda_sq=Re*Q*Density_ratio*St*St;
  }
```

In the interest of brevity we omit the listings of the assignments for the other cases. Finally, we select the length of the time interval over which the influx is ramped up from zero to its maximum value to be equal to 20 timesteps, create a constitutive equation for the solid, and document the parameter values used in the simulation:

```
  // Ramp period (20 timesteps)
  Ramp_period=Dt*20.0;

  // "Big G" Linear constitutive equations:
  Constitutive_law_pt = new GeneralisedHookean(&Nu,&E);

  // Doc
  oomph_info « std::endl;
  oomph_info « "-------------------------------------------"
             « std::endl;
  oomph_info « "Case: " « case_id « std::endl;
  oomph_info « "Re            = " « Re « std::endl;
  oomph_info « "St            = " « St « std::endl;
  oomph_info « "ReSt          = " « ReSt « std::endl;
  oomph_info « "Q             = " « Q « std::endl;
  oomph_info « "Dt            = " « Dt « std::endl;
  oomph_info « "Ramp_period   = " « Ramp_period « std::endl;
  oomph_info « "Max_flux      = " « Max_flux « std::endl;
  oomph_info « "Density_ratio = " « Density_ratio « std::endl;
  oomph_info « "Lambda_sq     = " « Lambda_sq « std::endl;
  oomph_info « "Gravity       = " « Gravity « std::endl;
  oomph_info « "Ignore fluid  = " « Ignore_fluid_loading« std::endl;
  oomph_info « "-------------------------------------------"
             « std::endl « std::endl;
 }

}// end_of_namespace
```

1.6 The driver code

The driver code has the usual structure, though in this case we use the command line arguments to indicate which case (FSI1, FSI2, FSI3, CSM1 or CSM2) to run. The absence of a command line argument is interpreted as the code being run as part of `oomph-lib's` self-test procedure in which case we perform a computation with the parameter values for case FSI1 and perform only a few timesteps.

```
//=======start_of_main==================================================
/// Driver
//======================================================================
int main(int argc, char* argv[])
{
 // Store command line arguments
 CommandLineArgs::setup(argc,argv);

 // Get case id as string
 string case_id="FSI1";
 if (CommandLineArgs::Argc==1)
  {
   oomph_info « "No command line arguments; running self-test FSI1"
              « std::endl;
  }
 else if (CommandLineArgs::Argc==2)
  {
   case_id=CommandLineArgs::Argv[1];
  }
 else
  {
   oomph_info « "Wrong number of command line arguments" « std::endl;
   oomph_info « "Enter none (for default) or one (namely the case id"
              « std::endl;
   oomph_info « "which should be one of: FSI1, FSI2, FSI3, CSM1"
              « std::endl;
  }
 std::cout « "Running case " « case_id « std::endl;
```

We set up the global parameter values, create a `DocInfo` object and trace file to record the output, and build the problem.

```
 // Setup parameters for case identified by command line
 // argument
 Global_Parameters::set_parameters(case_id);

 // Prepare output
 DocInfo doc_info;
 ofstream trace_file;
 doc_info.set_directory("RESLT");
 trace_file.open("RESLT/trace.dat");
 // Length and height of domain
 double length=25.0;
 double height=4.1;

 //Set up the problem
 TurekProblem<AlgebraicElement<RefineableQTaylorHoodElement<2> >,
  RefineableQPVDElement<2,3> > problem(length, height);
```

Next, we choose the number of timesteps (using a smaller number for a validation run, and for the case FSI1 in which the system rapidly approaches a steady state) and initialise the time-stepping for an impulsive start from the zero flow solution.

```
 // Default number of timesteps
 unsigned nstep=4000;
 if (Global_Parameters::Case_ID=="FSI1")
  {
   std::cout « "Reducing number of steps for FSI1 " « std::endl;
   nstep=400;
  }

 if (CommandLineArgs::Argc==1)
  {
   std::cout « "Reducing number of steps for validation " « std::endl;
   nstep=2;
  }

 //Timestep:
 double dt=Global_Parameters::Dt;

 // Initialise timestep
 problem.initialise_dt(dt);

 // Impulsive start
 problem.assign_initial_values_impulsive(dt);
```

Finally, we document the initial condition and start the time-stepping procedure, setting the `first` flag to `false` because we have not specified an analytical expression for the initial conditions that could be re-assigned after the

mesh adaptation when computing the first timestep.

```
 // Doc the initial condition
 problem.doc_solution(doc_info,trace_file);
 doc_info.number()++;
 // Don't re-set the initial conditions when adapting during first
 // timestep
 bool first = false;
 // Max number of adaptation for time-stepping
 unsigned max_adapt=1;
 for(unsigned i=0;i<nstep;i++)
  {
  // Solve the problem
  problem.unsteady_newton_solve(dt,max_adapt,first);

  // Output the solution
  problem.doc_solution(doc_info,trace_file);

  // Step number
  doc_info.number()++;
 }
 trace_file.close();

}//end of main
```

1.7 The Problem class

The `Problem` class contains the usual member functions, such as access functions to the various meshes. Because the nodal positions are updated by an algebraic node-update procedure, the function `actions_before_newton_convergence_check()` is employed to update the nodal positions in response to changes in the (solid) variables during the Newton iteration. The function `actions_before_implicit_timestep()` is used to adjust the influx during the start-up period.

```
//====start_of_problem_class=========================================
/// Problem class
//====================================================================
template< class FLUID_ELEMENT,class SOLID_ELEMENT >
class TurekProblem : public Problem
{

public:

 /// Constructor: Pass length and height of domain
 TurekProblem(const double &length, const double &height);

 /// Access function for the fluid mesh
 RefineableAlgebraicCylinderWithFlagMesh<FLUID_ELEMENT>* fluid_mesh_pt()
  { return Fluid_mesh_pt;}

 /// Access function for the solid mesh
 ElasticRefineableRectangularQuadMesh<SOLID_ELEMENT>*& solid_mesh_pt()
  {return Solid_mesh_pt;}

 /// Access function for the i-th mesh of FSI traction elements
 SolidMesh*& traction_mesh_pt(const unsigned& i)
  {return Traction_mesh_pt[i];}

 /// Actions after adapt: Re-setup the fsi lookup scheme
 void actions_after_adapt();

 /// Doc the solution
 void doc_solution(DocInfo& doc_info, ofstream& trace_file);

 /// Update function (empty)
 void actions_after_newton_solve() {}

 /// Update function (empty)
 void actions_before_newton_solve(){}

 /// Update the (dependent) fluid node positions following the
 /// update of the solid variables before performing Newton convergence
 /// check
 void actions_before_newton_convergence_check();

 /// Update the time-dependent influx
 void actions_before_implicit_timestep();

private:

 /// Create FSI traction elements
 void create_fsi_traction_elements();

 /// Pointer to solid mesh
 ElasticRefineableRectangularQuadMesh<SOLID_ELEMENT>* Solid_mesh_pt;
```

```
 /// Pointer to fluid mesh
 RefineableAlgebraicCylinderWithFlagMesh<FLUID_ELEMENT>* Fluid_mesh_pt;

 /// Vector of pointers to mesh of FSI traction elements
 Vector<SolidMesh*> Traction_mesh_pt;

 /// Combined mesh of traction elements -- only used for documentation
 SolidMesh* Combined_traction_mesh_pt;

 /// Overall height of domain
 double Domain_height;

 /// Overall length of domain
 double Domain_length;

 /// Pointer to solid control node
 Node* Solid_control_node_pt;

 /// Pointer to fluid control node
 Node* Fluid_control_node_pt;
};// end_of_problem_class
```

1.8 The problem constructor

We start by building the solid mesh, using an initial discretisation with 20 x 2 elements in the x- and y-directions. (The length of the flag is determined such that it emanates from its intersection with the cylinder and ends at x=6; The `origin` vector shifts the "lower left" vertex of the solid mesh so that its centreline is aligned with the cylinder.)

```
//=====start_of_constructor=============================================
/// Constructor: Pass length and height of domain
//======================================================================
template< class FLUID_ELEMENT,class SOLID_ELEMENT >
TurekProblem<FLUID_ELEMENT,SOLID_ELEMENT>::
TurekProblem(const double &length,
             const double &height) :  Domain_height(height),
                                      Domain_length(length)
{
 // Increase max. number of iterations in Newton solver to
 // accomodate possible poor initial guesses
 Max_newton_iterations=20;
 Max_residuals=1.0e4;

 // Build solid mesh
 //------------------

 // # of elements in x-direction
 unsigned n_x=20;

 // # of elements in y-direction
 unsigned n_y=2;

 // Domain length in y-direction
 double l_y=Global_Parameters::H;

 // Create the flag timestepper (consistent with BDF<2> for fluid)
 Newmark<2>* flag_time_stepper_pt=new Newmark<2>;
 add_time_stepper_pt(flag_time_stepper_pt);

 /// Left point on centreline of flag so that the top and bottom
 /// vertices merge with the cylinder.
 Vector<double> origin(2);
 origin[0]=Global_Parameters::Centre_x+
  Global_Parameters::Radius*
  sqrt(1.0-Global_Parameters::H*Global_Parameters::H/
       (4.0*Global_Parameters::Radius*Global_Parameters::Radius));
 origin[1]=Global_Parameters::Centre_y-0.5*l_y;

 // Set length of flag so that endpoint actually stretches all the
 // way to x=6:
 double l_x=6.0-origin[0];

 //Now create the mesh
 solid_mesh_pt() = new ElasticRefineableRectangularQuadMesh<SOLID_ELEMENT>(
  n_x,n_y,l_x,l_y,origin,flag_time_stepper_pt);
```

We create an error estimator for the solid mesh and identify a control node at the tip of the flag to track its motion.

```
 // Set error estimator for the solid mesh
 Z2ErrorEstimator* solid_error_estimator_pt=new Z2ErrorEstimator;
 solid_mesh_pt()->spatial_error_estimator_pt()=solid_error_estimator_pt;


 // Element that contains the control point
```

```
FiniteElement* el_pt=solid_mesh_pt()->finite_element_pt(n_x*n_y/2-1);

// How many nodes does it have?
unsigned nnod=el_pt->nnode();

// Get the control node
Solid_control_node_pt=el_pt->node_pt(nnod-1);

std::cout « "Coordinates of solid control point "
          « Solid_control_node_pt->x(0) « " "
          « Solid_control_node_pt->x(1) « " " « std::endl;
```

Finally, we perform one uniform mesh refinement and disable any further mesh adaptation.

```
// Refine the mesh uniformly
solid_mesh_pt()->refine_uniformly();

//Do not allow the solid mesh to be refined again
solid_mesh_pt()->disable_adaptation();
```

Next, we attach `FSISolidTractionElements` to the boundaries of the solid mesh that are exposed to the fluid. We complete their build by specifying which boundary of the bulk mesh they are attached to, as this information is required when setting up the fluid-structure interaction; see Further comments and exercises.

```
// Build mesh of solid traction elements that apply the fluid
//-------------------------------------------------------------
// traction to the solid elements
//-------------------------------

// Create storage for Meshes of FSI traction elements at the bottom
// top and left boundaries of the flag
Traction_mesh_pt.resize(3);
// Now construct the traction element meshes
Traction_mesh_pt[0]=new SolidMesh;
Traction_mesh_pt[1]=new SolidMesh;
Traction_mesh_pt[2]=new SolidMesh;

// Build the FSI traction elements
create_fsi_traction_elements();

// Loop over traction elements, pass the FSI parameter and tell them
// the boundary number in the bulk solid mesh -- this is required so
// they can get access to the boundary coordinates!
for (unsigned bound=0;bound<3;bound++)
 {
  unsigned n_face_element = Traction_mesh_pt[bound]->nelement();
  for(unsigned e=0;e<n_face_element;e++)
   {
    //Cast the element pointer and specify boundary number
    FSISolidTractionElement<SOLID_ELEMENT,2>* elem_pt=
    dynamic_cast<FSISolidTractionElement<SOLID_ELEMENT,2>*>
     (Traction_mesh_pt[bound]->element_pt(e));

    // Specify boundary number
    elem_pt->set_boundary_number_in_bulk_mesh(bound);

    // Function that specifies the load ratios
    elem_pt->q_pt() = &Global_Parameters::Q;

   }
 } // build of FSISolidTractionElements is complete
```

Finally, we create `GeomObject` representations of the three surface meshes of `FSISolidTraction↩Elements`. We will use these to represent the curvilinear, moving boundaries of the fluid mesh.

```
// Turn the three meshes of FSI traction elements into compound
// geometric objects (one Lagrangian, two Eulerian coordinates)
// that determine the boundary of the fluid mesh
MeshAsGeomObject*
 bottom_flag_pt=
 new MeshAsGeomObject
 (Traction_mesh_pt[0]);
MeshAsGeomObject* tip_flag_pt=
 new MeshAsGeomObject
 (Traction_mesh_pt[1]);
MeshAsGeomObject* top_flag_pt=
 new MeshAsGeomObject
 (Traction_mesh_pt[2]);
```

The final mesh to be built is the fluid mesh whose constructor requires pointers to the four `GeomObjects` that represent the cylinder and three fluid-loaded faces of the flag, respectively. We represent the cylinder by a `Circle` object:

```
// Build fluid mesh
```

```
//-----------------

//Create a new Circle object as the central cylinder
Circle* cylinder_pt = new Circle(Global_Parameters::Centre_x,
                                 Global_Parameters::Centre_y,
                                 Global_Parameters::Radius);
```

We build the mesh and identify a control node (a node at the upstream face of the cylinder), before creating an error estimator and performing one uniform mesh refinement.

```
// Allocate the fluid timestepper
BDF<2>* fluid_time_stepper_pt=new BDF<2>;
add_time_stepper_pt(fluid_time_stepper_pt);
// Build fluid mesh
Fluid_mesh_pt=
 new RefineableAlgebraicCylinderWithFlagMesh<FLUID_ELEMENT>
 (cylinder_pt,
  top_flag_pt,
  bottom_flag_pt,
  tip_flag_pt,
  length, height,
  l_x,Global_Parameters::H,
  Global_Parameters::Centre_x,
  Global_Parameters::Centre_y,
  Global_Parameters::Radius,
  fluid_time_stepper_pt);

// I happen to have found out by inspection that
// node 5 in the hand-coded fluid mesh is at the
// upstream tip of the cylinder
Fluid_control_node_pt=Fluid_mesh_pt->node_pt(5);

// Set error estimator for the fluid mesh
Z2ErrorEstimator* fluid_error_estimator_pt=new Z2ErrorEstimator;
fluid_mesh_pt()->spatial_error_estimator_pt()=fluid_error_estimator_pt;

// Refine uniformly
Fluid_mesh_pt->refine_uniformly();
```

We now add the various meshes to the `Problem's` collection of sub-meshes and combine them to a global mesh

```
// Build combined global mesh
//---------------------------

// Add Solid mesh the problem's collection of submeshes
add_sub_mesh(solid_mesh_pt());

// Add traction sub-meshes
for (unsigned i=0;i<3;i++)
 {
  add_sub_mesh(traction_mesh_pt(i));
 }

// Add fluid mesh
add_sub_mesh(fluid_mesh_pt());
// Build combined "global" mesh
build_global_mesh();
```

The application of boundary conditions for the solid are straightforward: All displacements of the flag's left end (mesh boundary 3) are suppressed; the other faces are free. Strictly speaking, the pinning of the redundant solid pressure nodes is superfluous since the `RefineableQPVDElement` used for the discretisation of the flag employ a displacement-based formulation, but it is good practise to perform this step anyway to "future-proof" the code for the use of other element types.

```
// Apply solid boundary conditons
//-------------------------------
//Solid mesh: Pin the left boundary (boundary 3) in both directions
unsigned n_side = mesh_pt()->nboundary_node(3);
//Loop over the nodes
for(unsigned i=0;i<n_side;i++)
 {
  solid_mesh_pt()->boundary_node_pt(3,i)->pin_position(0);
  solid_mesh_pt()->boundary_node_pt(3,i)->pin_position(1);
 }
// Pin the redundant solid pressures (if any)
PVDEquationsBase<2>::pin_redundant_nodal_solid_pressures(
 solid_mesh_pt()->element_pt());
```

The fluid has Dirichlet boundary conditions (prescribed velocity) everywhere apart from the outflow where only the horizontal velocity is unknown.

```
// Apply fluid boundary conditions
//--------------------------------
//Fluid mesh: Horizontal, traction-free outflow; pinned elsewhere
unsigned num_bound = fluid_mesh_pt()->nboundary();
```

```
for(unsigned ibound=0;ibound<num_bound;ibound++)
 {
  unsigned num_nod= fluid_mesh_pt()->nboundary_node(ibound);
  for (unsigned inod=0;inod<num_nod;inod++)
   {
    // Parallel, axially traction free outflow at downstream end
    if (ibound != 1)
     {
      fluid_mesh_pt()->boundary_node_pt(ibound,inod)->pin(0);
      fluid_mesh_pt()->boundary_node_pt(ibound,inod)->pin(1);
     }
    else
     {
      fluid_mesh_pt()->boundary_node_pt(ibound,inod)->pin(1);
     }
   }
 }//end_of_pin
// Pin redundant pressure dofs in fluid mesh
RefineableNavierStokesEquations<2>::
 pin_redundant_nodal_pressures(fluid_mesh_pt()->element_pt());
```

We impose a parabolic inflow profile with the current value of the influx at the inlet (fluid mesh boundary 3).

```
// Apply boundary conditions for fluid
//------------------------------------

// Impose parabolic flow along boundary 3
// Current flow rate
double t=0.0;
double ampl=Global_Parameters::flux(t);
unsigned ibound=3;
unsigned num_nod= Fluid_mesh_pt->nboundary_node(ibound);
for (unsigned inod=0;inod<num_nod;inod++)
 {
  double ycoord = Fluid_mesh_pt->boundary_node_pt(ibound,inod)->x(1);
  double uy = ampl*6.0*ycoord/Domain_height*(1.0-ycoord/Domain_height);
  Fluid_mesh_pt->boundary_node_pt(ibound,inod)->set_value(0,uy);
  Fluid_mesh_pt->boundary_node_pt(ibound,inod)->set_value(1,0.0);
 }
```

We complete the build of the solid elements by passing them the pointer to the constitutive equation, the gravity vector and the timescale ratio:

```
// Complete build of solid elements
//--------------------------------

//Pass problem parameters to solid elements
unsigned  n_element =solid_mesh_pt()->nelement();
for(unsigned i=0;i<n_element;i++)
 {
  //Cast to a solid element
  SOLID_ELEMENT *el_pt = dynamic_cast<SOLID_ELEMENT*>(
   solid_mesh_pt()->element_pt(i));

  // Set the constitutive law
  el_pt->constitutive_law_pt() =
   Global_Parameters::Constitutive_law_pt;

  //Set the body force
  el_pt->body_force_fct_pt() = Global_Parameters::gravity;

  // Timescale ratio for solid
  el_pt->lambda_sq_pt() = &Global_Parameters::Lambda_sq;
 }
```

The fluid elements require pointers to the Reynolds and Womersley (product of Reynolds and Strouhal) numbers:

```
// Complete build of fluid elements
//--------------------------------

// Set physical parameters in the fluid mesh
unsigned nelem=fluid_mesh_pt()->nelement();
for (unsigned e=0;e<nelem;e++)
 {
  // Upcast from GeneralisedElement to the present element
  FLUID_ELEMENT* el_pt = dynamic_cast<FLUID_ELEMENT*>
   (fluid_mesh_pt()->element_pt(e));

  //Set the Reynolds number
  el_pt->re_pt() = &Global_Parameters::Re;

  //Set the Womersley number
  el_pt->re_st_pt() = &Global_Parameters::ReSt;

 }//end_of_loop
```

Setting up the fluid-structure interaction is done from "both" sides" of the fluid-solid interface: First we en-

sure that the no-slip condition is automatically applied to all fluid nodes that are located on the three faces of the flag (mesh boundaries 5, 6 and 7). This is done by passing the function pointer to the
 \c FSI_functions::apply_no_slip_on_moving_wall() function to the relevant fluid nodes (<A href="../../../navier_stokes/osc_ellipse/html/index.html">recall</A> that the auxiliary node update functions are automatically executed whenever the position of a node is updated by the algebraic node update). Since the no-slip condition (1) involves the Strouhal number (which, in the current problem, is not equal to the default value of \c FSI_functions::Strouhal_for_no_slip=1.0), we overwrite the default assignment with the actual Strouhal number in the problem. \until // done automatic application of no-slip Next, we set up the lookup schemes required by the \c FSISolidTractionElements to establish which fluid elements affect the traction onto the solid: \until } All interactions have now been specified and we conclude by assigning the equation numbers \skipline // Assign equation numbers \until //end_of_constructor <HR> <HR> @section set_traction Create traction elements This is a helper function that attaches \c FSISolidTractionElement to the solid elements that are exposed to the fluid traction. We store the elements in three distinct sub-meshes – one for each face. (Yet another mesh, pointed to by \c Combined_traction_mesh_pt, is created for post-processing purposes.) \skipline start_of_create_traction_elements \until // end of create_traction_elements <HR> <HR> @section check Actions before Newton convergence check The algebraic node-update procedure updates the positions in response to changes in the solid displacements but this is not done automatically when the Newton solver updates the solid mechanics degrees of freedom. We therefore force a node-update before the Newton convergence check. \dontinclude turek_flag.cc \skipline start_of_actions_before_newton_convergence_check \until } <HR> <HR> @section timestep Actions before the timestep Before each timestep we update the inflow profile for all fluid nodes on mesh boundary 3. \skipline start_of_actions_before_implicit_timestep \until end_of_actions_before_implicit_timestep <HR> <HR> @section after_adapt Actions after adapt After each adaptation, we unpin and re-pin all redundant pressures degrees of freedom. This is necessary because their "redundant-ness" may have been altered by changes in the refinement pattern; see <A href="../../../navier_stokes/adaptive_driven_cavity/html/index.html"> another tutorial</A> for details. We ensure the automatic application of the no-slip condition on fluid nodes that are located on the faces of the flag, and re-setup the FSI lookup scheme that tells \c FSISolidTractionElements which fluid elements are located next to their Gauss points. \skipline actions_after_adapt \until end of actions_after_adapt <HR> <HR> @section doc Post-processing The function \c doc_solution(...) produces the output for the fluid, solid and traction meshes and writes selected data to the trace file. \skipline start_of_doc_solution \until end_of_doc_solution <HR> <HR> @section com_ex Further comments and exercises - When completing the build of the \c FSISolidTractionElements (the elements that apply the fluid traction to the solid elements that are exposed to the fluid) we specified the number of the solid mesh boundary they are located on, using \n\n @code elem_pt->set_boundary_number_in_bulk_mesh(bound); \endcode \n This information is required when setting up the fluid-structure interaction because the \c MeshAsGeomObject representation of the mesh of \c FSISolidTractionElements is parametrised by the boundary coordinate in the solid mesh. Explore the details of the implementation by commenting out the relevant line of code and use the debugger to find out how and where the code fails. \b Note: Since this step is somewhat subtle and therefore easily forgotten, the \c FSISolidTractionElements issue an explicit warning if the bulk boundary number has not been set – but only if the the library is compiled in PARANOID mode. \n\n - When comparing our results against those in Turek & Hron's benchmark, we only focused on the period and amplitude of the fully-developed self-excited oscillations. The benchmark data also provides data on the time-dependent variations of the drag and lift coefficients. Design suitable \c FaceElements (to be attached to the faces of the Navier-Stokes elements that are adjacent to the flag or the cylinder) to compute these quantities. The <a href="../../../the_data_structure/html/classoomph_1_1NavierStokesSurfacePowerElement.html"> <code>NavierStokesSurfacePowerElements</code></a> should provide a good basis for these. <HR> <HR> @section ackn Acknowledgements - This code was originally developed by Stefan Kollmannsberger and his students Iason Papaioannou and
 Orkun Oezkan Doenmez. It was completed by Floraine Cordier. <HR> <HR> @section sources Source files for this tutorial - The source files for this tutorial are located in the directory:\n\n <CENTER> <A href="../../../../demo_drivers/interaction/turek_flag/"> demo_drivers/interaction/turek_flag/ </A> </CENTER>\n - The driver code is: \n\n <CENTER> <A href="../../../../demo_drivers/interaction/turek_flag/turek_flag.cc"> demo_drivers/interaction/turek_flag/turek_flag.cc </A> </CENTER> <hr> <hr> @section pdf PDF file A <a href="../latex/refman.pdf">pdf version of this document is available. \